UNIVERSITY OF CAMBRIDGE
Judge Business School
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The Implications of AI and AI Adoption in Real Estate

GCP 2025

BlackRock

Agenda

- Introduction
- Evolution and Future of AI
- Major Trends Related to AI
- AI's Opportunities in Real Estate Investment
- Reconstructing a Structured AI Judgment Process
- Risks of Using AI and How to Mitigate Them
- Recommendations



01 Introduction



Project Background | This project aims to provide BlackRock **strategic recommendations** on effectively leveraging AI for Real Estate investments

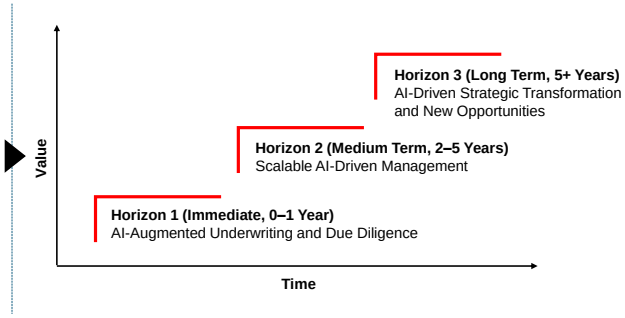
Key Research Questions

1. AI's Influence on Real Estate **Demand & Human Behaviour**
2. **AI in Investment Strategy**
3. **Risks & Challenges**

Executive Summary | We aim to provide a roadmap of AI adoption for BlackRock

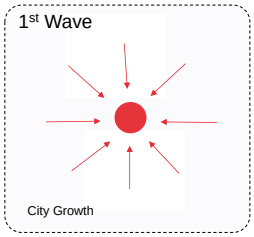
BlackRock Key Competitive Advantages

- Scale and Data Access
- Risk Management Expertise
- Long-Term Capital & Patient Strategy
- Technological Adoption and Innovation
- Global Diversification and Network

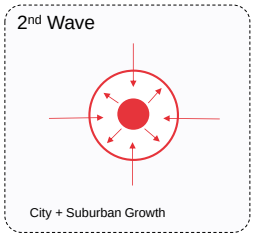


RE Through Waves | Tracking real estate performance across historical markets to spot investment patterns

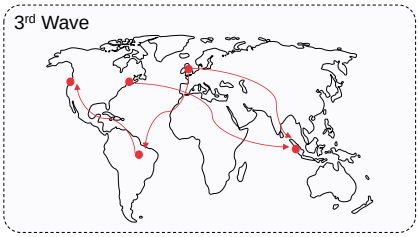
Eras & Real Estate Evolution



Industrial Revolution
Industrialisation triggered rapid urban growth. Industrial became a new asset class.



Post-WW2 Mechanisation
Highlights suburbanisation and infrastructure-led RE growth.



Computer & Information Age
Tech boom & globalisation drove urban resurgence & cross-border investments in knowledge hubs.

Covid Era
Remote work & e-commerce accelerated decentralisation & logistics.

AI: Fourth Wave | Tech and AI are reshaping real estate's demand, liquidity and exit models

Key Lessons
Disruption → Opportunity

Tech Reshapes Geography: Reshoring Trend
Winners & Losers
Anticipate Policy Reforms
People's Preferences Evolve

The Dual Edge of AI

Boon

Institutional Investors: Sharper, faster, more predictive

Retail Segment: More informed yet emotion-driven

Bane

Weakened public finances → Drop in Retail buyers → **Liquidity shrinks** → Trapped or discounted assets (indirect effect)

Rise of rentals Build-to-rent replaces build-to-sell
Need for shift in **Exit Strategy**



02 Evolution and Future of AI



AI – Artificial Intelligence | Traditional and Generative AI

Traditional AI – Intelligence exhibit by machines

Machine Learning – Algorithmic based learning

Deep Learning – Complex Neural Networks

Generative AI

- Deep learning algorithms generating human-like content.
- Generative means deriving content from large set of other samples.
- Flexible on how to query data.

Stages of AI

Stage 1 – ANI
Specialised AI in one area

Stage 2 – AGI
As smarts as a human across the board

Stage 3 – ASI
Much smarter than the best human in every field

Dimension	Traditional AI	Generative AI
Core Function	**Task-oriented**: Focuses on analysing data, making predictions, and optimising based on predefined rules	**Creation-oriented**: Creates new content (text, images, audio, code) based on learned patterns
Techniques Used	Employs algorithms like decision trees, support vector machines, convolutional neural networks, and recurrent neural networks	Utilses neural network techniques like transformers, Generative Adversarial Networks (GANs), and Variational Autoencoders (VAEs)
Data Processing	Excels with structured data; analyses without generating new data	Processes both structured and unstructured data; generates entirely new outputs
Example	Recommendation systems, facial recognition, search algorithms	Text generators (ChatGPT), image creators (DALL-E), music composers, code generators
Ethical Concerns	Algorithmic bias, privacy issues, and transparency challenges	Creation of deepfakes, misinformation, copyright concerns, and content authenticity issues
Best used for	**Solutions, prediction, classification and optimisation**	**Content creation, possibility exploration**

Future of AI | AI will be smarter and cheaper fuelled by the advancement in computing power with exponential growth and step-changes

Emerging Trends

Continued Advancement of Moore's Law
Despite repeated speculations about the end of Moore's Law, semiconductor leaders like Intel and TSMC continue to develop smaller and more efficient chips.

Emerging Alternative Computing Architectures
Silicon photonics – using light rather than electricity to carry data – represents one of the most promising approaches. There is also a shift toward memory-centric computing.

Rise of Quantum Computing
Quantum AI promises breakthroughs in Optimisation and learning speed

Sustainability and Energy Efficiency
Next-gen AI models are being trained and run with lower carbon footprints

How Are Advancements Trending?

Traditional AI – Considerable improvement in accuracy by leveraging massive amounts of data.

Generative AI – Huge enhancements in abilities but hallucination will remain.

Direct Impacts

Human Workforce
- AI automation of high-value, structured tasks
- Workflows reimagined: AI-centric processes
- Human labour: transformation, not extinction
- Jobs lost; jobs created – sectoral impacts
- Decrease in information asymmetry

Required Infrastructure
- Energy infrastructure
- Decentralised datacentres

03 AI Forecast: By Asset Classes



Front Runners | AI is directly accelerating demand in select asset classes

Data centres

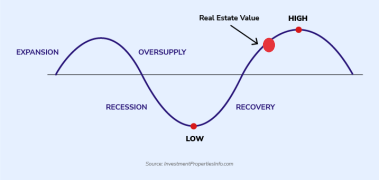
- Strategic backbone of the AI economy.
- Rising digital infrastructure needs are driving sustained demand from global tech firms.
- Trend of shift from mega datacentres to **smaller, modular datacentres** at the edge of networks.

Logistics & Warehousing

- Facilities that support AI-enabled distribution are in strong demand.
- Focus on **last-mile hubs**, **automated distribution centres** and well-connected logistics corridors to mitigate oversupply risk and ensure stable income.

Life Sciences & Healthcare

- AI is fuelling demand for R&D labs, biotech spaces and high-spec medical facilities.
- Hot market especially near university clusters and healthcare ecosystems.
- **Exit Uncertainty:** Lab spaces require heavy capex and specialised infrastructure.



Opportunity

- Focus on **flexible design** - the kind that can adapt to future expansion, technological shifts or operational iterations.
- Closely tracking how **demand and supply are evolving**, especially in high-growth niches.

Marginal Gainers | The net effect is positive and evolutionary

Living Sector

SFH & MFH
- AI integration enhances **operational efficiency** via smart integrated systems, predictive maintenance and automated leasing efficiencies.
- Potential for **modest NOI uplift** through energy savings and lower vacancy.
- Boom in "secondary markets" and built to rent models.

Student Housing
- AI tools support dynamic pricing, occupancy forecasting and tenant screening; enhancing revenue optimisation and management efficiency.
- Demand remains **linked to education cycles and international mobility**, not directly AI-led.

Senior Living
- AI could enable more personalised care and smart monitoring, enhancing service.
- Likely to see **gradual demand uplift** from tech-savvy aging populations but driven more by **demographic trends** than AI itself.

Hotels & Hospitality

- AI adoption in dynamic pricing, personalisation and operational automation supports margin enhancement.
- Guest experience and brand differentiation improve, but demand remains **cyclical and macro-sensitive**.

Opportunity

BRAINBOX AI

- AI can ↓ Expenses → **Increased NOI** (↑) → **Boost IRR**
- **AI-powered energy management systems** can cut utility costs by optimising heating, lighting, and water usage in real time.

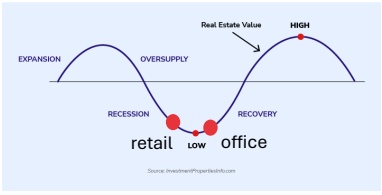
Stragglers | The asset classes facing structural headwinds in AI-driven economy

Office

- Pressure from remote and hybrid work models, with AI further reducing the need for physical presence by streamlining knowledge work.
- AI-enhanced virtual collaboration tools reduce the necessity of colocation, especially for transactional or back-office functions.
- Demand is polarising, with premium spaces thriving while lower-quality offices struggle.

Retail

- AI is accelerating the shift to e-commerce through advanced personalisation, automation and delivery efficiencies.
- Survivors will be experiential and tech-enabled, often requiring fewer but higher-performing physical locations.
- Older retail centres risk becoming stranded assets unless reimagined with clear integration of tech, data and services.



Opportunity

wework 2.0

- Focus on **adaptive reuse** - the kind that can adapt to people's demand and preferences.
- Look for Flex Work Ecosystems – invest in or retrofit for flexible office platforms.

Rising Trend | A Shift in The Relationship Between Landlords and Tenants

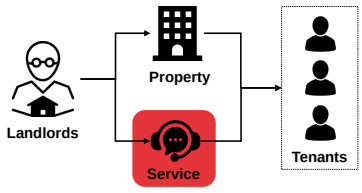
There is a major trend towards *service-oriented* real estate.

Buildings will become more intelligent and will be managed more like a user-centric product.

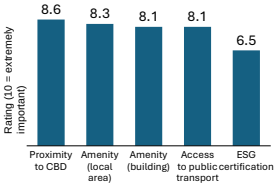
AI is accelerating this trend to deliver a higher quality of service and value to tenants.

Owners who see themselves as landlords, and not as service providers, risk being left behind as the tenants require more responsive spaces.

As an investor, the implication is that an emphasis on asset management, with a tenant-service focus, will be key to managing a successful portfolio.



Behind location, what a building can offer is the most important consideration in office space



What are factors people considers when looking property?

04 AI's Opportunities in Real Estate Investment



Data is Key | The better the data, the better the AI

Type of Data	Proprietary	Social Media and Sentiment	Open	Academic
Data source	Transaction records	• Social platforms (Twitter/X, Facebook) • Crowdsourced sites (Yelp, Reddit)	Government portals (Data.gov)	Papers on forecasting commercial RE trends
Possible AI Application	• More accurate price estimates • Automated detection of anomalous or missing data	Sentiment and trend analysis to anticipate micro-neighbourhood risks	• Cross-checking proprietary and social data • Benchmarking local/regional real estate trends	In-depth risk modelling (flood zones, climate impact, social shifts)
Challenges	• Strict confidentiality (PII) • Bias from incomplete coverage	• Bias in user groups (not fully representative) • Privacy issues around scraping or storing user-generated data	• Format inconsistencies across jurisdictions • Data fragmentation	Mismatch between academic findings and industry constraints

Data Processing and Integration | AI can help create a 360° view of an asset or market

Large and varied datasets bring technical challenges, including:

Data quality and standardisation

Data sparsity or errors

Privacy and access restrictions add complexity

Volume of data

AI in Data Processing

Collecting

Natural Language Processing for Unstructured Data
Extracts insights from text-based property descriptions into data points.

Cleansing

Machine Learning for Anomaly Detection
Spot unusual property values or transaction prices that may indicate errors or fraud.

Clustering for Data Consistency
Algorithms to find inconsistencies and streamline data cleaning in large datasets.

Geospatial AI for Location Data Cleaning
Fix location-related errors, while geo-statistics handle missing coordinates.

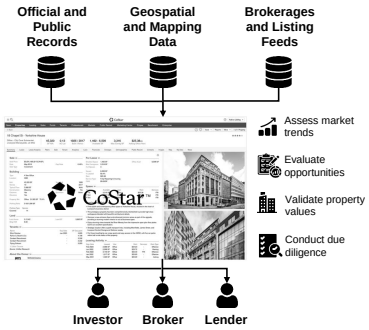
Fulfilling

Predictive modelling for Missing Data
AI models predict missing values based on existing patterns.

Generative AI for Data Simulation
Can create synthetic transaction data to compensate for when transaction records are limited.

Data Integration Use Case | CoStar

CoStar is a leading commercial real estate data integration and analytics platform



Benefits of CoStar's Integrated Data Platform

Centralised Information
All critical property details and market insights are housed in one place.

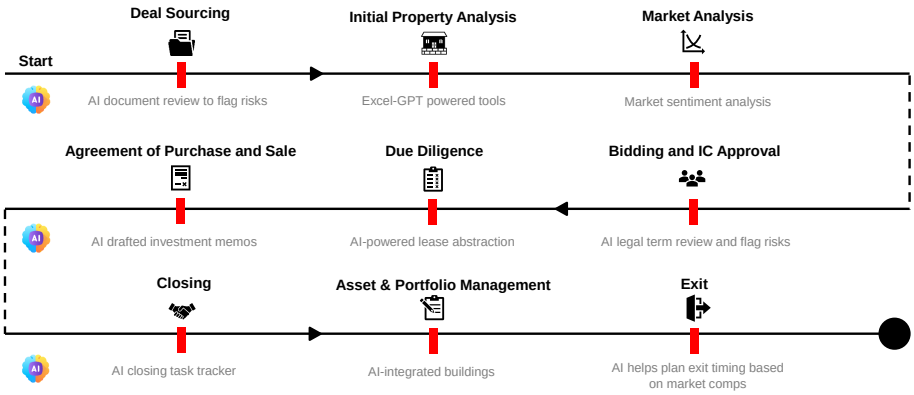
Improved Data Accuracy and Consistency
Consolidated property records ensure a more reliable dataset for analysis.

Enhanced Market Visibility
Real-time updates offer greater transparency into market trends.

Time and Resource Savings
Streamlined data collection and cleaning reduces manual effort.

Scalable Analytics and Reporting
Data quality supports advanced analytics.

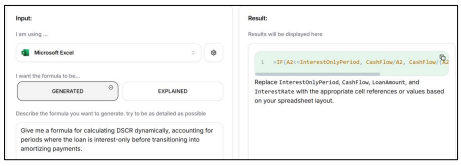
AI Opportunities in a Sample Real Estate Investment Workflow



Efficiency In Underwriting | GPT Excel - A tool that generates and explains Excel formulas from plain text.

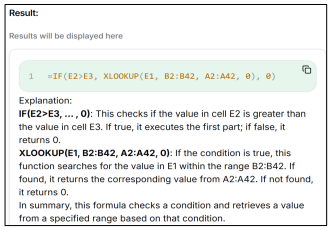
Example 1

Using GPT Excel to adjust a DSCR formula. No VBA or excel skills are needed, just a plain language input. This saves time and reduces training required.



Example 2

Using GPT Excel to explain a complex formula found in a model.



Preliminary Deal Analysis | Market Sentiment Analysis

Market Sentiment Analysis

AI is used to analyses **news, social media and other media** to detect trends and tone changes across assets and regions.

Tools have been developed to track big data sets such as consumer reviews to **identify changing tenant preferences and neighbourhood sentiment**.

This use-case of AI **delivers early signals of market momentum** to support smarter investment decisions ahead of lagging indicators like leasing or transaction data.

Case Study

REDFIN

Redfin recognised the need to enhance its market analysis capabilities beyond traditional data sources.

They developed an AI-enabled **market sentiment analysis tool to analyse customer reviews**.

AI sentiment analysis enabled Redfin to interpret emotional tone and market trends from diverse textual sources, offering early indicators of market shifts before traditional data reflects them.

Efficiency In Due Diligence | Lease Abstraction

Lease Abstraction

A review of lease documentation is an essential part of any due diligence process.

Machine learning models have been trained on thousands of lease agreements and provide accurate data, cost savings and efficient lease abstractions.

Traditionally this was done by lawyers, and **often took days, but now can be done in hours.**

Case Study

Prophia

Seattle-based URG (14M sf portfolio) used Prophia in their recent acquisition of a retail mall.

The AI-powered lease abstraction software was **used to review 100 tenant leases in a 10-day DD period**.

The **process quickly identified 20+ documents** that were missing and critical questions that URG needed resolved by the seller.

The lease abstracts provided by Prophia were beneficial in expediting the review of Argus files, enhancing the valuation and cash flow forecasting process.

Case Studies | Enhancing Sustainable Real Estate Investing Through AI in Valuations

CoreLogic, Australia

Automated Valuation Models
AI driven AVMs were used to process vast datasets swiftly, resulting in more accurate property valuations. **Their AVM Explorer tool allows users to delve into factors influencing a property's estimated value,** enhancing confidence in valuations. This reduces the need for **redundant site visits** and physical inspections **while optimising land usage.**

Enhancing Property Data Analytics
Collaborations with companies like Haystacks.AI demonstrate CoreLogic's commitment to providing comprehensive property data solutions. By supplying nationwide data directly to clients' cloud environments, CoreLogic simplifies data integration, allowing for scalable analytics and informed investment decisions.

CoreLogic



Norway's SWF

Norges Bank Investment Management (NBIM), which manages Norway's sovereign wealth fund, has been integrating artificial intelligence (AI) into its investment strategies to enhance capital deployment and operational efficiency.

Key Outcomes
- NBIM uses AI to enhance capital deployment and efficiency.
- AI **reduces unnecessary trades** and optimises portfolio adjustments.
- Targeted **10% productivity boost** via AI integration within 12 months.
- OpenAI CEO suggested potential **20% productivity improvement.**
- **Reduction of operational waste and energy** supporting long-term investment in sustainable assets.
- NBIM urges portfolio companies to adopt responsible AI practices.
- Emphasizes AI transparency, oversight, and clear purpose in sensitive sectors.
- Major real estate investments in **UK (Regent St.)** and **US properties.**
- AI likely informs NBIM's long-term real estate investment strategy.

Norges Bank Investment Management



Case Studies | Enhancing Sustainable Real Estate Investing Through AI in Infrastructure

Canada Pension Investment Board

Investing in AI Infrastructure
Canadian pension funds are investing in AI infrastructure, recognising the importance of data centres. In October 2024, CPP Investments and Singapore's GIC partnered with Equinix in a **$15 billion venture to expand U.S. hyperscale data centres, supporting AI capabilities** and influencing real estate investment decisions. Equinix next-gen data centers are designed to be **energy-efficient and support sustainable growth.**

Key Outcomes
- Network enables private machine learning between asset managers.
- Support **better investment decisions across multiple asset classes.**
- AI infrastructure investments may influence real estate strategies.
- Pension funds exploring generative AI for investment Optimisation.
- AI adoption expected to **enhance real estate portfolio management.**

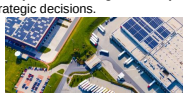


JLL

JLL GPT proprietary LLM for RE markets and Infrastructure
Empowers employees and clients with instant, data-driven answers about real estate markets, investment opportunities, infrastructure trends**,** and property data. Provides stakeholders with instant insights into **sustainable building materials, energy usage trends, and carbon footprint data**.

Training Data: Internal real estate market data, proprietary research reports and historical property transaction data

Key Outcomes
- Due Diligence Support: Speed up document reviews and extraction of relevant information.
- Client Advisory: Enable **real estate brokers and advisors to respond faster and more accurately.**
- Internal Productivity: **Reduce time spent searching and compiling data,** allowing teams to focus on strategic decisions.



Risk Analysis | AI-driven analytics are enabling earlier identification and quantification of a wide range of risks in real estate portfolios.

Advanced machine learning (ML), predictive modelling, computer vision, and NLP techniques transform raw data into forward-looking risk indicators. These tools can rapidly synthesise multi-factor risk profiles for properties, investors, or markets, far beyond what manual analysis could achieve.

Predictive Analytics for Market & Financial Risk
- AI models forecast **market volatility, prices and occupancy** by analysing historical and real-time data.
- During the pandemic, **machine learning flagged shifts in transaction volumes and listing inventories.**

Property Valuation and Depreciation Risk
- Modern **automated valuation models (AVMs),** such as HouseCanary and Zillow's Zestimate**.**
- Predict long-term depreciation or appreciation by analysing remote work trends or tech-sector growth.
- Computer vision tools detect deterioration, for example, **ZestyAI's "Roof Age**" feature identifies roof condition to avert unplanned repair costs.

Zillow

Environmental and Climate Risk
- Geospatial analytics platforms (e.g., Cape Analytics) **merge satellite data with climate models to highlight flood, wildfire, or hurricane risks.**
- AI also **evaluates properties' energy usage and emissions** to anticipate regulatory penalties, thus guiding ESG-conscious strategies.

CAPE ANALYTICS

Socio-Demographic Risk Factors
- Machine learning **incorporates demographic data, job growth, and population shifts to pinpoint rising or declining markets**.
- NLP algorithms parse local news and social media for **hints of neighbourhood change,** e.g., new amenities may signal gentrification, while frequent closures or crime reports suggest market decline.

Asset-Specific Risk
- NLP-driven document analysis reviews leases, titles, and other legal records, flagging unusual clauses or unresolved liens.
- Predictive models **simulate long-term performance under various assumptions including maintenance costs, tax changes and demographic shifts** to reveal likely returns and potential pitfalls.
- Early detection of risk factors helps avoid costly surprises post-acquisition.

05 Reconstructing a Structured AI Judgement Process



Thinking Outside the Black Box | Reconstruct a Structured AI Judgement Process

Redefining Explainability in AI-Driven Investment Decisions

The unsolvable AI black box problem: April 3rd, Anthropic published their latest research further confirming that existing AI explainability is a pseudo-concept in their groundbreaking paper "Reasoning Models Don't Always Say What They Think" .

Why does it matter

Because the Black Box issue Kills Trust

For BR:

For the same question:

1.AI gives you an answer---->How can I explain to investors why this answer is credible?
2. AI gives you another answer: How do I explain this inconsistency to investors?

For BR's clients:
1.Why should I choose/trust BR instead of your competitors?
2. Does your investment decision suggestions make sense to me? (If so, how)

For AI expert:
No trust without understanding. No scale without trust.

Rethinking Explainability

1. When we talk about explainability, what are we really talking about?

True Intentions:
Technical breakthrough? ❌
To create greater value in real-world business contexts? ✅

2. A thinking dilemma rather than a dead end :
A formalism trap caused by
"technological solutionism"

3. The truly unavoidable question :
What kind of rationale-backed AI decision-making process would produce answers that clients would actually buy into?



Reasoning Models Don't Always Say What They Think

Yanda Chen Joe Benton Ansh Radhakrishnan Jonathan Uesato Carson Denison
John Schulman* Arushi Somani
Peter Hase* Miles Wagner Fabien Roger Vlad Mikulik
Sam Bowman Jan Leike Jared Kaplan Ethan Perez

Alignment Science Team, Anthropic

Abstract

Chain-of-thought (CoT) offers a potential boon for AI safety as it allows monitoring a model's CoT to try to understand its intentions and reasoning processes. However, the effectiveness of such monitoring hinges on CoTs faithfully representing models' actual reasoning processes. We evaluate CoT faithfulness of state-of-the-art reasoning models across 6 reasoning hints presented in the prompts and find: (1) for most settings and models tested, CoTs reveal their use of hints at least 1% of the time in the majority of examples where they use the hint, but the reveal rate is often below 20%, (2) outcome-based reinforcement learning initially improves faithfulness but plateaus without saturating, and (3) when reinforcement learning increases how frequently hints are used (reward hacking), the propensity to verbalize them does not increase, even without training against a CoT monitor. These results suggest that CoT monitoring is a promising way of noticing undesired behaviors during training and evaluations, but that it is not sufficient to rule them out. They also suggest that in settings like ours where CoT reasoning is not necessary, test-time monitoring of CoTs is unlikely to reliably catch rare and catastrophic unexpected behaviors.

Redefining explainability

1. A Redefinition: **Explainability = Priceability**

2. A Practical Approach :

Design a verification and reasoning framework capable of assessing the reliability of outputs produced by large language models (LLMs)

E.g., the BR model ——— A proprietary LLM co-developed with OpenAI.

3. The system should incorporate the following features:

Explicit logic chain (VS) The black-box nature

Quantifiable risk assessment
(VS) The randomness and volatility

Built-in adaptability for future scenarios

Three-Stage Structural Intelligence | Reconstruct a Structured AI Judgement Process

Guided by this new orientation, a structural intelligence methodology emerges.



What is Structural Intelligence?

1.Not a single AI product, but a process: Designed to help LLM judgement more stable, more explainable, and to have transaction value for investment.

2. It has a **three-step** approach involving **two** types of AI models

3.Built on a foundation of data quality and scale——Two data-strengthening approaches are applied:

Micro-level
Macro-level



Three steps of structural intelligence I

1. Data augmentation

Uncovering Signals in Micro Realities ;
Utilise modern technologies to uncover and structure fragmented, up-to-date industry signals.

Macro-Future Simulation:
Synthetic data mitigates bias from limited historical datasets and enhances future-facing investment simulations.

What is synthetic data?

1. Training phase: Model / algorithm calibration, the expansion of training datasets

2. Validation phase: Stress testing + causal inference



3 steps of structural intelligence II

2. Hybrid AI Model System with Mutual Reinforcement

Internally customised LLM: BlackRock GPT
Core Value: Generate diverse and innovative investment strategies by integrating insights from unstructured data.

AI-Augmented Quantitative Models:
A deep integration of traditional quantitative methodologies with advanced AI techniques
Core Value:
Transforming black-box AI outputs into interpretable, actionable, and priceable financial products through structured validation.

3.Adaptive Execution & Closed-Loop Optimisation:
Powered by real-time monitoring
and data-driven feedback loops.

Demo: Datacentre Scenario as an Example | Reconstruct a Structured AI Judgement Process

Ask BR GPT questions

Json (well-structured output format)

```
{
  "Recommended Exit Strategy":
"REITs listing",
  "Key Assumptions": [
    "Electricity price annual growth ≤ 5%",
    "Data centre utilisation rate ≥ 80%",
    "Sufficient market liquidity for REITs"
  ],
  "Expected IRR": "15%"
}
```

AI-Augmented Quantitative Models: GPT Output Verification

Automatically decompose input into verifiable modules:

Assumption 1:
Annual electricity price growth ≤ 5%

Assumption 2:
Data centre utilisation rate ≥ 80%

Assumption 3:
Sufficient market liquidity for REITs

Conclusion: Expected IRR = 15%

Then conduct modular alignment of real and synthetic data to enable stress testing, historical backtesting, causal inference validation, and other forms of scenario-based testing.

Verify electricity price assumptions

Verify utilisation assumptions

Verify REITs liquidity assumptions

Verify expected IRR

Generate a validation report:

structured and visualised results

Feedback Loop: Validation Results Automatically Sent Back to GPT

Trigger Conditions:
Key assumption confidence < defined threshold (e.g., 60%)
Stress test and other tests failure

Optimisation Process:
Validation results from the structured AI model are fed back to GPT
GPT adjusts the output based on feedback

06 Risks of AI and Mitigations



Risk Assessment | Assessing Risks and Mitigation Actions on GenAI

Impact / Category	Risk for Real Estate Investor and Decision Maker	Reputational Risk for Real Estate Investment Company	Financial Risk for Real Estate Investment Company	Risk for Workforce of Real Estate Investment Company	Legal/Regulation Risk for Real Estate Investment Company
Minor	**Risk**: Overreliance on black-box AI outputs may mislead individual investors. **Mitigation**: Use AI tools with interpretable outputs and human-in-the-loop oversight.	**Risk**: Public scepticism if model outputs lack transparency. **Mitigation**: Publish model explainability reports.	**Risk**: Slight inefficiencies from suboptimal but explainable models. **Mitigation**: Balance performance with interpretability.	**Risk**: Confusion or lack of trust in AI tools. **Mitigation**: Provide training on explainable AI principles.	**Risk**: Non-compliance with soft guidelines. **Mitigation**: Monitor evolving explainability standards.
Moderate	**Risk**: Inaccurate intermediate outputs may lead to poor investment choices. **Mitigation**: Validate models using scenario testing and feature attribution tools.	**Risk**: Criticism from media or clients over opaque decision-making. **Mitigation**: Engage in stakeholder co-design and communications training.	**Risk**: Investment losses from feature attribution errors. **Mitigation**: Use reliable techniques (e.g., decision trees) and expert validation.	**Risk**: Job displacement due to misunderstood AI roles. **Mitigation**: Involve staff in AI adoption and re-skill accordingly.	**Risk**: Breaches of transparency obligations. **Mitigation**: Use interpretable models and log decision rationale.
High	**Risk**: Misinterpretation of generative AI insights could drive systemic misallocation of capital. **Mitigation**: Apply explainability techniques like counterfactuals and external auditing.	**Risk**: Public backlash from biased or unexplained model predictions. **Mitigation**: Incorporate diverse datasets and regular ethical reviews.	**Risk**: Major losses from reliance on hallucinated LLM outputs. **Mitigation**: Use multi-agent and retrieval-augmented systems to reduce error.	**Risk**: Workforce resistance or disengagement. **Mitigation**: Apply participatory design methods to integrate human expertise.	**Risk**: Legal penalties due to non-compliance with AI laws. **Mitigation**: Align AI development with regulatory frameworks and conduct legal audits.

Safeguarding Generative AI | A hybrid approach using rule-based and classifier ML guardrails for safe, accurate Responses

Real estate conversations must respect fair housing laws that prohibit discriminatory practices, such as “steering” potential buyers or renters toward or away from properties based on protected characteristics (race, religion, disability, etc.).

Pre-processor: Scans incoming user requests for potential non-compliance.

Post-processor: Validates the LLM’s generated response before passing it on to the user.

3 Core Approach - current best practice is to blend all three.

Prompt Engineering:
- Insert instructions in the system prompt that explicitly forbid discriminatory responses and remind the model to follow fair housing laws, using the RHODES Framework
- Advantage: Easy to implement, keeps many responses in compliance.
- Drawback: Can be overly cautious, mistakenly blocking legitimate requests (low precision).

Stop List:
- Maintain a list of high-severity, banned terms or phrases—if the model output or user input contains them, automatically refuse or filter the request.
- Advantage: Guarantees certain content never makes it through.
- Drawback: Lacks context—simple keyword matching can lead to errors (false positives).

Classifier Model:
- Build a specialised machine learning classifier (e.g., fine-tuned BERT) to detect potential fair housing violations in either the user’s query or the LLM’s response.
- Advantage: More nuanced and context-aware, tuneable precision/recall, and less likely to over-block.
- Drawback: Requires building and maintaining a labelled dataset, more complex engineering.

Prompt and Key Word Engineering | Best Practices

For best results with LLMs, be clear and specific, provide context and structure prompts logically. Ask for step-by-step reasoning, refine iteratively, and use examples to guide output. These practices help ensure more accurate, relevant and useful responses.

Provide Context

Give background, role, or audience details to help the model tailor responses more accurately to your needs.

Ask for Step-by-Step Reasoning

Encourage logical thinking by asking for step-by-step explanations, which improves accuracy and transparency in complex tasks.

Be Specific and Clear

Use precise language and direct instructions to reduce ambiguity and guide the model towards the desired outcome.

Use Iterative Refinement

Review, adjust, and clarify prompts based on output to progressively improve the quality and relevance of responses.

Use Structured Prompts

Organise requests using lists or sections to guide the model in producing well-formatted, easy-to-follow responses.

Leverage Few-Shot or Zero-Shot Examples

Show examples of ideal output to help the model learn and mimic the structure, tone, or format expected.

Successful Implementation of GenAI in Global Firm | Morgan Stanley Private ChatGPT

In 2023, Morgan Stanley implemented a private GPT-4 system for wealth management, enabling advisors to access internal insights and generate client-ready summaries securely, while maintaining compliance; setting a benchmark for secure, explainable GenAI use in financial investments.

Continuous Monitoring and Improvement

Feedback loops for users to rate outputs and report inconsistencies were stablished; ensuring model evolved with market conditions and internal strategy.

Domain-Specific Data for Training

The models were trained using high-quality, in-house data related to property markets, asset performance, and financial metrics. Incorporated region-specific legal and tax considerations.

Embedded Explainability Features

Explainability was integrated clarifying how the model arrives at suggestions. Summarisation prompts, source referencing, and visual aids were used to foster trust from stakeholders.

Defined Clear Use Cases and Limitations

The models' scope was limited to assistive tasks such as market commentary. It was avoided letting it make final investment calls without human review, and transparent guardrails.

Involved Investment Tuning Experts

Human-in-the-loop reinforcement learning (RLHF) involved experienced investment professionals to fine-tune outputs and decision logic with institutional expectations.

Data Privacy and Compliance

ChatGPT was setup a secure, private environment; either on-premises or via a trusted cloud partner; with full compliance with data protection regulations (e.g. GDPR, CCPA).

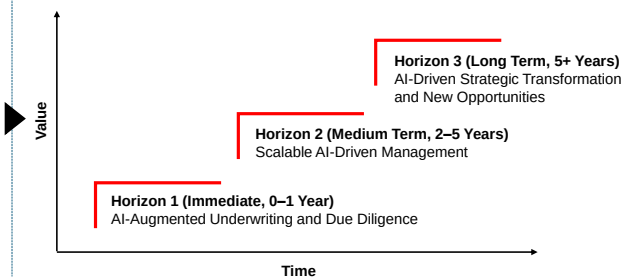
03 Recommendation



Recommendation | To tackle the uncertainty in AI, Three-horizon framework are applied to shape the roadmap of AI adoption for BlackRock

BlackRock Key Competitive Advantages

- Scale and Data Access
- Risk Management Expertise
- Long-Term Capital & Patient Strategy
- Technological Adoption and Innovation
- Global Diversification and Network



Horizon 1 (Immediate, 0–1 Year) | AI-Augmented Underwriting and Due Diligence

Establishing the foundation	**Enhance Data Infrastructure**	BlackRock should consolidate and clean its real estate data into a **unified platform,** creating a "CoStar-like" database of internal and external information.
Launch pilot projects	**Integrate AI in Deal Analysis**	Use AI algorithms in underwriting models to rapidly analyse deals and market data, for instance by verifying **property valuations** alongside traditional appraisals.
	Automate Due Diligence Tasks	Implement AI including **reading AI for time-consuming due diligence tasks like reviewing leases, contracts and financial statements.** By using proven software (as in the Prophia case), BlackRock can reduce time consuming in extracting key lease terms, flagging risks and identifying missing documents.
	Improve Valuation and Underwriting Accuracy	Use AI for **scenario analysis and risk flagging in underwriting**. Models can run thousands of simulations (varying assumptions like lease-up periods, interest rates, or cap rates) to identify downside risks and flag anomalies by comparing against past deals.
	Deploy AI Productivity Tools	Immediately equip investment teams with **AI-driven productivity tools,** such as natural language query interfaces (e.g., a chatGPT plugin) to quickly retrieve insights from BlackRock's databases. Also deploy interpretability tools like **SHAP and LIME** to ensure transparent predictions.

Horizon 2 (Medium Term, 2–5 Years) | Scalable AI-Driven Management

Strengthen the foundation	Foster Ecosystem	**Scale Data and Tech Infrastructure**	As AI expands, BlackRock must scale infrastructure in Horizon 2 through **cloud computing and robust governance**. Partnerships with cloud providers or data firms can feed continuous property, market, and ESG data into models. **Ensuring data privacy, model validation and ethical AI** by training on segmented data and using synthetic inputs keeps decisions compliant and fair.
		Build a Scalable AI Platform	BlackRock should establish an **enterprise-level AI platform** for real estate, following the lead of industry pioneers. This strategy could include creating a proprietary Real Estate LLM (Large Language Model) in **partnership** with OpenAI and using **co-design methodologies** to involve both staff and clients in model validation and interpretation. Such an approach would build stakeholder trust and **foster a robust AI ecosystem**.
		Talent and Culture Development	By the medium term, BlackRock should have a **dedicated AI-focused real estate team and training programs** to upskill traditional professionals on AI outputs. **Fostering a collaborative culture** where AI is seen as a partner, rather than a threat, will help maximise its benefits.
		Adopt the Structural Intelligence Framework	BlackRock can pilot the structural intelligence approach in underwriting and portfolio strategy, ensuring **every investment decision is backed by an explainable AI** "decision chain" that highlights both macro factors (GDP growth, AI adoption rate, interest rates) and micro factors (occupancy, energy efficiency, etc.).
Extend application		**AI-Driven Portfolio Management**	With foundational tools in place, Horizon 2 shifts to portfolio-level optimisation. BlackRock can use AI to **rebalance portfolios, optimise allocations and detect concentration risks or opportunities** (e.g., tenant-type correlations with tech stocks). Real-time data feeds enable **dynamic adjustments**.

Horizon 3 (Long Term, 5+ Years) | AI-Driven Strategic Transformation and New Opportunities

Investing in AI-Centric Assets and Infrastructure	By this stage, AI will reshape real estate demand, prompting BlackRock to **tilt its portfolio toward digital infrastructure** (data centres, cloud facilities, telecom sites). AI will also **boost demand for life sciences labs and tech incubator spaces**, so BlackRock should ensure adequate exposure to these sectors.
Embed AI Fully in Strategic and Investment Decision-Making	BlackRock can create **advanced AI forecasting models with synthetic data to imagine future states**. Instead of standard 10-year proformas, a Horizon-3 approach might use a perpetual **digital twin,** fed by real data and AI-generated scenarios (economic, demographic, climate), to continuously simulate outcomes. Beyond forecasting, agentic GenAI can support **multi-source due diligence** and **personalise investor updates**.
New Investment and Ownership Models	AI might also enable fundamentally new ways to invest in and own real estate, and BlackRock should be prepared to innovate here. We might see the rise of **tokenised real estate platforms** where AI facilitates fractional ownership and liquidity, or AI-driven marketplaces matching global investors to assets in real-time.
Aligning with Emerging Urban Infrastructure Needs	By aligning investments with the **infrastructure of the future (digital, transportation, and energy)**, BlackRock ensures its portfolio remains relevant and valuable as cities modernise in the AI era.

Q&A



Thank You



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or inferred.

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Interviews | Dr. Victor Romero-Cano

- **Direct Perception for Interpretability**: Dr. Romero uses *intermediate representations* (e.g., scene segmentation) in end-to-end models to improve transparency while maintaining optimisation.

- **Stakeholder-Friendly Design**: He highlights the need for model outputs and reasoning to be *clear to non-technical stakeholders*, especially in high-impact decisions.

- **Feature Importance via Decision Trees**: Prefers decision trees due to their *built-in feature ranking*, useful in fields with well-known influential variables.

- **Limits of Generative AI**: Notes that while LLMs offer insights via attention, they often *overlook critical details*. Favour models that output meaningful representations.

- **Bias Mitigation through Data Diversity**: Emphasises *data quality and variety* as key to reducing bias, beyond just developing new model architectures.

- **Minimising Hallucinations in LLMs**: Supports *multi-agent setups* and *search-augmented generation (RAG)* to reduce hallucinated outputs and enhance reliability.

- **Human-in-the-Loop and Co-Design**: Advocates for *participatory design*, engaging stakeholders throughout development to improve trust and usability.

Interviews | Nikodem Szumilo

•**AI's Role in Trends**: AI may not accelerate trends but will amplify value in service-oriented, well-managed assets; early adopters of AI in asset research gain first-mover advantage.
•**Valuations & Preferences**: AI enables a value-add strategy by turning around mismanaged assets; success depends on operational execution and access to real-time data and tools.
•**Tool Adoption Strategy**: Avoid building custom GPTs—use public tools (e.g., ChatGPT) with strong internal training; custom models quickly become obsolete.
•**Risk Management**: AI helps spot overpriced assets; useful in illiquid or volatile markets (e.g., office).
•**Bias & Privacy**: AI is biased but potentially less so than humans; bias must be acknowledged, and training is key. Data privacy concerns are solvable with IT resources.
•**Real-World Use Cases**:
•*Hilton*: AI-driven revenue management led to higher returns.
•*Maersk*: GenAI-enabled mass negotiation increased portfolio value.
•**Challenges to Adoption**:
•Skill gaps & lack of understanding.
•Overestimating data/privacy concerns.
•Uncertainty about practical implementation.
•**Other Tech (Blockchain)**: Not transformative; AI has deeper and broader impact.

Interviews | Antony Slumbers

•**AI Cost vs. Capability**: Intelligence is getting cheaper and better—AI will take over more cognitive tasks, requiring RE to evolve alongside rising machine capability.
•**Two AI Types**:
•*Predictive AI*: Structured, repeatable tasks.
•*Generative AI*: Creative tasks; inherently less accurate.
Human value lies in tasks AI can't yet replicate.
•**Sector Impacts**:
•*Office*: AI strengthens remote/hybrid work trends—premium on quality space and purposeful design.
•*Retail*: Shift toward experiential and logistics-driven formats.
•*Residential*: Undersupplied; AI reshapes location/form factors (modular builds, secondary cities, tech-enabled units).
•*Industrial*: Faster, tech-enabled, localized; potential reuse of obsolete offices.
•*Data Centres*: Growth depends on power infra; trend toward localized "data-centre-in-a-box" models.
•**AI in Underwriting**:
•Challenges: RE lacks deep datasets; difficult to apply traditional ML at scale.
•Opportunities: Use GenAI to simulate or augment datasets; boost deal analysis, documentation, risk assessment.
•**Physical Asset Optimization**:
•Sensor-rich buildings + AI = "MRI for buildings"; leads to higher valuation & optimized user experience.
•**AI Tools in Use**:
•*Laiout.co* for floorplan design.
•Emphasis on user-centered, operationally excellent assets—not just financial engineering.
•**Strategy Shift**:
•RE becomes more productized; asset management must deliver great experience.
•Adapt portfolios to hyper-growth sectors; anticipate winners and losers in the AI era.

Interviews | Thies Lindenthal

•**AI & Adjacent Tech**:
•Blockchain: Overhyped; limited current value for real estate beyond possible smart contracts.
•**AI Explainability & Risk**:
•BlackRock must make AI outputs interpretable, especially in investment contexts.
•GenAI's opacity is a legal risk in sensitive decisions (e.g., hiring, eviction). Safer for asset investment.
•Transparency and model understanding are key to trust and compliance.
•**Consumer Behavior & Demand Shifts**:
•4 drivers of change:
1. Economic/productivity shifts → location/value creation changes
2. Asset productivity → more services, better UX, higher intensity
3. Market frictions reduced (liquidity, info asymmetry, transaction costs)
4. AI as tenant (data centers, infra)

•**Valuation & Location**:
•Tech improvements reduce need for office proximity (e.g., real-time translation, high-quality remote interaction).
•Location preferences will shift as remote tools improve.
•**Asset Class Impacts**:
•AI companies increasing demand for high-spec RE (e.g., data centers with robust infra).
•Bigger impact on location, design, infrastructure needs.
•**AI in Underwriting & Risk**:
•Widespread adoption (e.g., JLL GPT) helps firms unlock internal data.
•Still requires high-quality data and clear risk protocols.
•**Operational Optimization**:
•Predictive AI aids planning and efficiency. Human broker role becomes more relational and strategic.
•Fewer junior roles—AI as industrial revolution for white-collar RE.
•**Promising Tools**:
•*Viability* (development feasibility), *Deafly* (Netherlands), climate risk tools, insurance sector rapidly adopting AI.
•**Risks & Mitigation**:
•"Garbage in, garbage out"—data quality is crucial. Must reduce bias by better understanding users and design impact.
•AI can close the feedback loop: use-space insights to inform future RE design.

AI in Built Environment |

AI Application	Benefits to Landlords/Managers	Benefits to Tenants/Occupants	Example Platforms/Cases
1. Predictive Maintenance AI monitors building equipment to predict and prevent failures	– Reduces unexpected breakdowns and downtime (improved service continuity) – Lowers repair and replacement costs by fixing issues early – Extends lifespan of HVAC, elevators, etc.	– Fewer disruptions (e.g. heating or elevator always available) – Safer and more reliable environment (less risk of system failures) – Consistently comfortable conditions (no sudden outages)	**R8 Digital Operator** (R8 Technologies)– AI HVAC maintenance in offices **ServiceChannel** – automates maintenance workflows for FM teams
2. Energy Optimization AI dynamically adjusts HVAC, lighting, and power use	– Cuts energy costs and improves building's sustainability metrics – Optimizes energy use 24/7 beyond human capabilities (granular control) – Can earn green building certifications or incentives	– Comfortable indoor climate is maintained more consistently (AI avoids overcooling/overheating) – Occupants contribute to sustainability (lower carbon footprint building) – Potential cost savings passed on in utilities or rent	**BrainBox AI** – retrofits HVAC AI (25% energy cost reduction, 40% GHG cut) **The Edge (Amsterdam)** – AI-driven BMS for lighting/HVAC (highly energy-efficient)
3. Occupancy Analytics Sensors and AI analyze space usage and occupancy patterns	– Data-driven space planning (e.g. consolidate underused areas to save cost) – Informs lease strategies and flexibility (offer right-sized spaces) – Optimizes cleaning, staffing, and HVAC schedules based on actual use	– Workspaces that match actual needs (not too crowded or too empty) – Better availability of shared resources (rooms, desks) due to optimized allocation – Healthier environment (e.g. AI avoids over-crowding, ensures air quality)	**IBM TRIRIGA AI** – provides real-time space utilization insights **Beringar** – IoT sensors + AI for occupancy and air quality monitoring

AI in Built Environment |

AI Application	Benefits to Landlords/Managers	Benefits to Tenants/Occupants	Example Platforms/Cases
4. Digital Twins Live digital replica of the building for monitoring and simulation	– Central platform to monitor all building systems in real time – Predictive "what-if" analysis (simulate changes or failures virtually) – Improves coordination among teams (single source of truth for building data) – Achieved ~20% energy and ~25% maintenance cost reductions in cases	– Quicker response to issues (building operators see problems in the twin immediately) – More reliable and optimized building performance (twin anticipates needs) – Can enable rich 3D visuals for occupants (e.g. navigation or VR experiences of the building)	**Willow** – digital twin platform used in commercial portfolios (e.g. Brookfield Properties) **Sensgreen** – offers digital twin solutions (reports 20% energy savings)sensgreen.com **Microsoft Azure Digital Twins** – used for large campus management
5. Smart Building Automation AI algorithms autonomously control HVAC, lighting, elevators, etc. in an integrated way	– Hands-off operations: routine adjustments made automatically (saves management time) – Optimizes multiple systems together for overall efficiency (holistic management) – Ensures consistently optimal settings, improving asset performance (equipment not overrun or underutilized)	– Environment adapts in real-time to occupant needs (no manual intervention required for comfort changes) – Higher indoor comfort and wellness (AI maintains temperature, air quality, lighting at ideal levels) – Quick resolution of issues (system self-corrects or alerts staff before tenants notice)	**Siemens Desigo w/ AI** – intelligent BMS adjusting all building systems **Schneider Electric EcoStruxure** – uses AI for energy, lighting, security control **Johnson Controls OpenBlue** – autonomous building platform (e.g. in smart hospitals)
6. Tenant Experience AI Apps, chatbots, and services tailored to occupants	– Differentiates the property with premium services (attracts/retains tenants) – Collects data on tenant preferences and pain points (feedback for improvements) – Automates customer service tasks (lower staffing needs for front desk/basic inquiries)	– Convenience of on-demand services (book amenities, report issues, get info instantly) – Personalized content/offers (building news, events, or perks relevant to them) – Greater engagement and satisfaction, feeling "heard" by management	**HqO / VTS Rise / Equiem** – tenant engagement platforms with AI features **Coldwell Banker Commercial example** – AI chatbot handling tenant inquiries 24/7 **Zendesk + NLP** – used in facilities for instant query resolution

AI in Built Environment |

AI Application	Benefits to Landlords/Managers	Benefits to Tenants/Occupants	Example Platforms/Cases
7. Personalized Comfort AI learns individual or group preferences for climate and lighting	– Increases tenant satisfaction (happy tenants more likely to renew leases) – Can zone spaces more efficiently when knowing usage patterns (energy benefit too) – Provides a selling point: "our building adapts to you" marketing advantage	– Customizable workspace or home environment (lighting, temperature, even background music tailored to preference) – Improved wellness and productivity (people perform better when comfortable) – Sense of control over environment without needing to fiddle with controls constantly	**Comfy App (Siemens)** – lets office tenants request instant comfort changes **Smart Spaces** – UK platform emphasizing personalisation in offices **Lutron Athena** – uses AI for personalized lighting scenes in smart homes
8. AI Security & Access AI-enhanced surveillance and access control	– Enhances security with less manpower (AI monitors cameras at scale) – Rapid threat detection and response (improves risk management for the asset) – Modernizes building image (showing commitment to tenant safety)	– Faster entry/exit (e.g. facial recognition entry, no queues or repetitive ID checks) – Greater safety and peace of mind (security incidents identified and addressed faster) – Non-intrusive experience (security running in background)	**Kastle Systems** – AI-based building access (some offices use face recognition) **Envoy + Vision** – AI camera analytics for visitor management **Dense CCTV analytics** (multiple vendors use AI to flag anomalies in real time)

AI in Built Environment – Case Study | Laiout

Laiout is an AI-powered platform designed to streamline and automate interior space planning and layout design. From a high-level perspective, here is the core use case it addresses:

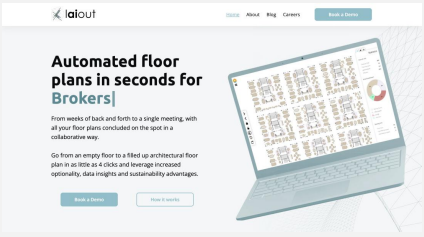
1.Automated Floor Plan Generation
Laiout uses algorithms and AI-driven tools to produce initial layout solutions—such as office, retail, or residential space configurations—quickly and with minimal manual effort.

2.Rapid Iterations
Instead of drafting plans by hand or going through multiple rounds of manual revisions, Laiout allows designers to experiment with different layouts rapidly. Users can update constraints (e.g., desired capacity, adjacency requirements) and immediately see alternative designs.

3.Time and Cost Savings
By reducing the time needed to test or refine a floor plan, Laiout significantly speeds up the design process. This can help businesses cut costs and move forward with final designs more quickly.

4.Data-Driven Decision Making
The platform typically leverages AI to account for practical considerations (like building codes or occupant comfort), which means layout suggestions are rooted in relevant data rather than just aesthetic preferences.

5.Enhanced Collaboration
Since designs can be quickly updated and shared, teams—whether architects, real estate developers, or interior designers—can give and receive feedback in near-real time, resulting in a smoother planning cycle.



In essence, **Laiout** solves a common bottleneck in the interior design and architectural process: manually generating, reviewing, and revising floor plans. By offering fast, AI-guided layouts and easy iteration, it helps professionals focus on the creative and strategic aspects of space design rather than spending excessive time on mechanical drafting

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